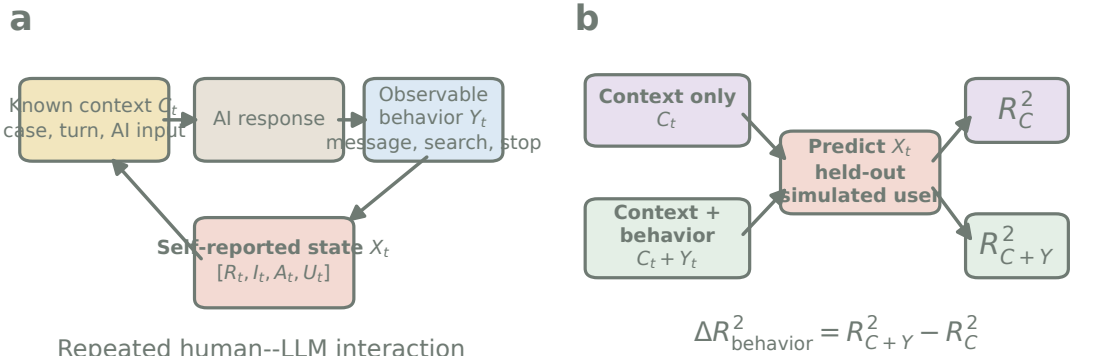

a
Known context C_t
case, turn, AI input
AI response
Observable behavior Y_t
message, search, stop
Self-reported state X_t
$[R_t, I_t, A_t, U_t]$
Repeated human--LLM interaction
b
Context only
C_t
Context + behavior
$C_t + Y_t$
Predict X_t
held-out simulated user
R^2_C
R^2_{C+Y}
$\Delta R^2_{\text{behavior}} = R^2_{C+Y} - R^2_C$